

Collin Zoeller



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EDUCATION

Carnegie Mellon University (non-degree)

Jul 2023 - Present

Relevant Coursework:

Pittsburgh, PA

- Intro to Machine Learning (10-601, Graduate)
- Quantum Integer Programming and Machine Learning (Graduate)

Brigham Young University

Apr 2023

Bachelor of Science: Economics (Emphasis in Econometrics)

Provo, UT

Minors: Mathematics, Spanish

GPA: 3.71

Relevant Coursework:

- Advanced Econometrics, Advanced Macroeconomics, Advanced Linear Algebra
- Machine Learning for Economists

WORKING PAPERS AND RESEARCH PROJECTS

Efficient Large-Scale Text Classification with ANN (Sole Author; white paper + implementation)

- Develop a scalable pipeline for clustering large text corpora using ANNOY + SLM
- Demonstrates improved computational efficiency for feature engineering over standard clustering
- Authored technical paper and an open-source implementation publicly available on GitHub

GAMA Quantum Optimizations for High-Dimensional Least Squares (Sole Author; Working Paper)

- Studies conditions for which the GAMA hybrid-quantum optimization solves the OLS problem
- Examines theory and implements it in D-wave simulated annealers

The Big Book of Estimators (Sole Author; Working Paper)

- Conspectus of estimators, assumptions, and use cases
- Intended for non-specialized researchers as a non-textbook reference for model selection

Artificial Intelligence in Economic Research: Applications, Promises, Pitfalls (Sole Author; in progress)

- Meta-analysis examining how applications of AI can harm and help researchers
- Produced slides and paper for future presentation
- Intended for general use and non-specialized researchers in the social sciences

RESEARCH EXPERIENCE

Tepper School of Business, Carnegie Mellon University

Jul 2023- Present

Pre-doctoral Research Associate

Pittsburgh, PA

- Lead IRS administrative data project (~1.5B observations) involving high-dimensional econometrics
- Build pipelines for data engineering: SQL data lake, file storage, project workflow (SQL, Linux, Stata)
- Develop ANN/NLP methods for extracting structure from 1B+ open-text fields (Python)
- Technical training and assistance to 3 other RAs in Linux, Python, Stata, and computational optimization

Harvard Business School

May 2022-Jul 2023

Computer Vision Programmer and Research Assistant for HBS Ph.D. Candidate

Cambridge, MA /Provo, UT

- Constructed custom image segmentation + OCR + text-recognition for dissertation using 25k+ images
- Extracted, and structured data from 1M+ data points for downstream economic and ML analysis
- Provided econometric design support and generalized OCR tools for new research applications

Brigham Young University Record Linking Lab

Jun 2022- Jul 2023

Research Assistant, Computer Vision Team Lead

Provo, UT

- Managed 6+ CV projects for FamilySearch and historical data partners (Python, AWS, CV/DL)
- Delivered key insights and research to support genealogical data transcription using CV techniques

CONTRIBUTIONS (RA WORK)

Evolution of the Relationship between the Gig Economy and Entrepreneurship: The Heterogeneous Effects of Labor Market Disruptions

Research Associate with authors Denes, Lagaras, and Tsoutsoura (2024)

- Estimate DiD using IRS taxpayer microdata to study entrepreneurial responses to labor disruptions
- Built ANN text clustering pipeline for 1+ billion deduction fields
- Constructed a framework to parallelize Stata (Python/Stata; precursor to `StataHelper`)
- Provided data engineering, replication, and publication support

Entrepreneurship and the Gig Economy: Evidence from U.S. Tax Returns, *Journal of Financial Economics*

Research Associate with authors Denes, Lagaras, and Tsoutsoura (2023)

- Estimate DiD using IRS microdata to identify entrepreneurial transitions among platform workers
- Cleaned and structured datasets, maintained empirical codebase for journal submission

First Come, First Served: The Timing of Government Support and Its Impact on Firms

Research Associate with authors Denes, Lagaras, and Tsoutsoura (2023)

- Studies small businesses receiving paycheck protection loans during the COVID-19 pandemic
- Estimate DiD effects of PPP timing on firm financial health using credit bureau data
- Data cleaning, workflows, and publication preparation

Take-up of Flexible Labor (work in progress)

Research Associate with authors Denes, Lagaras, and Tsoutsoura

- Studies employment of flexible labor through tax deduction claims
- trained text classifier RNN on 1 billion+ taxpayer-inputted deductions

PRESENTATIONS AND CONFERENCES

BYU President's Leadership Council

Sept 2022

Growing Together: Growing the Tree

Provo, UT

- Presented research and methodological implementations of computer vision in genealogical research

Family History Technology Workshop

Feb 2023

City Directories Automated Indexing

Provo, UT

- Presented custom OCR/NLP pipeline extracting data from city directories for economic research

TEACHING EXPERIENCE

Brigham Young University Economics Department

Sep 2021-Apr 2022

Econ 110 Teaching Assistant

Provo, UT

- Taught 500+ students in introductory macro and microeconomics
- Hosted 15 lab hours each week to students and 1-hour lecture review for exam preparation
- Crafted test questions and graded all assignments for the instructor

COMPUTATIONAL TOOLS & RESEARCH SOFTWARE

'StataAgent' Agentic AI tool for Natural Language Stata Interfacing (in progress)

Present

Author

Pittsburgh, PA

- Manipulates and analyzes data within Stata from user-inputted questions
- Useful for simple tasks (i.e. summary statistics) when correct syntax is confusing or not understood

'Explain' Stata Package for LLM Interfacing

February 2025

Author

Pittsburgh, PA

- Interfaces Stata with LLMs for code debugging, explanation, and error analysis
- Useful for learning Stata and for commenting and digesting code
- Open-source availability on GitHub

'StataHelper' Python Package for PyStata Parallelization

August 2024

Author

Pittsburgh, PA

- Systematically parallelizes variable Stata processes (simulations, tests, etc.)
- Wraps PyStata API and bridges syntax for simpler implementation
- Open-source availability on PyPi and GitHub, discussion on Medium

TECHNICAL SKILLS

Stata, Python, R, MATLAB, SQL, Linux, LaTeX, ArcGIS, AWS, Google SDK, Git/GitHub, Quantum ML, Quantum DL, Quantum Annealing